Akhand Bharat Academy
Class 10 Mathematics - Chapter 04
Quadratic Equations: Worked Examples

1. Standard form
   Every quadratic equation can be written as ax^2 + bx + c = 0, where a != 0.

2. Solving by factorisation
   Example: x^2 - 5x + 6 = 0
   Split the middle term: x^2 - 3x - 2x + 6 = 0
   Group: x(x - 3) - 2(x - 3) = 0
   So (x - 2)(x - 3) = 0, giving x = 2 or x = 3.

3. Completing the square
   Example: x^2 + 6x + 5 = 0
   x^2 + 6x = -5
   x^2 + 6x + 9 = 4
   (x + 3)^2 = 4, so x + 3 = +/- 2, giving x = -1 or x = -5.

4. The quadratic formula
   x = (-b +/- sqrt(b^2 - 4ac)) / 2a
   The discriminant D = b^2 - 4ac tells you the nature of the roots:
   D > 0 two distinct real roots
   D = 0 two equal real roots
   D < 0 no real roots

5. Board pattern questions
   These appeared in the 2023 and 2024 Bihar Board papers.
   Practise each one before the weekly test.